

Gabrielle Kaili-May Liu

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Curriculum Vita

Education

Yale University, 2023–Present
Ph.D. in Computer Science
Advisor: Arman Cohan
Anticipated Graduation: 06/28

Massachusetts Institute of Technology, 2019–2023
S.B. in Mathematics with Computer Science
S.B. in Brain and Cognitive Sciences
GPA: 5.0/5.0

Research Interests

LLM Uncertainty/Calibration, LLM Post-Training, LLM Evaluation, Synthetic Data Generation, Metacognition in LLMs

Availability

June 1, 2026–September 1, 2026

Professional Experience

Summer Camp for Applied Language Exploration (SCALE), Human Language Technology Center of Excellence (HLTCOE), Johns Hopkins University, Baltimore, MD, USA

Visiting Research Scholar, June 2025–August 2025

- Developed pipeline to generate challenging, uncheatable, multihop queries which are unanswerable over any given database, for use in adaptively benchmarking the refusal capabilities of leading retrieval augmented generation (RAG) systems.

Explorer of Perception and Attention (EPA) Laboratory, National Taiwan University, Taipei, ROC Taiwan

Visiting Researcher, June 2023–August 2023, June 2020–August 2021

- Established the first experimental procedure to reliably assess and quantify metacognition of emotion.

Collaborative, Learning, and Adaptive Robots (CLeAR) Lab, National University of Singapore, Singapore

Visiting Researcher, June 2022–August 2022

- Investigated trustworthy AI principles for agentic decision-making in evolving environments.

Publications

Li, B., Walden, W., Hou, Y., **Liu, G.**, Lawrie, D., Mayfield, J., Yang, E., Callison-Burch, C., Dietz, L. DoGMatIQ: Automated Generation of Question-and-Answer Nuggets for Report Evaluation. Submission to SIGIR 2026.

Liu, G., Li, B., Cohan, A., Walden, W., Yang, E. Investigating Retrieval-Augmented Generation Systems on Unanswerable, Uncheatable, Realistic, Multi-hop Queries. ECIR 2026.
<https://www.arxiv.org/abs/2510.11956>

Dietz, L., Li, B., **Liu, G.**, Ju, J.-H., Yang, E. Lawrie, D., Walden, W., Mayfield, J. Incorporating Q&A Nuggets into Retrieval-Augmented Generation. ECIR 2026.
<https://www.arxiv.org/abs/2601.13222>

Walden, W., Mason, M., Weller, O., Dietz, L., Conroy, J., Molino, N., Recknor, H., Li, B., **Liu, G.**, Hou, Y., Lawrie, D., Mayfield, J., Yang, E. Auto-ARGUE: LLM-Based Report Generation Evaluation. Submission to SIGIR 2026.
<https://arxiv.org/abs/2509.26184>

Liu, G., Yona, G., Caciularu, A., Szpektor, I., Rudner, T. G. J., Cohan, A. MetaFaith: Faithful Natural Language Uncertainty Expression in LLMs. EMNLP Main 2025.
<https://aclanthology.org/2025.emnlp-main.1505>

Bean, A. M., Kearns, R. O., Romanou, A., Hafner, F. S., Mayne, H., Batzner, J., Foroutan, N., Schmitz, C., Korgul, K., Batra, H., Deb, O., Beharry, E., Emde, C., Foster, T., Gausen, A., Grandury, M., Han, S., Hofmann, V., Ibrahim, L., Kim, H., Kirk, H. R., Lin, F., **Liu, G.**, Luettgau, L., Magomere, J., Rystrom, J., Sotnikova, A., Yang, Y., Zhao, Y., Bibi, A., Bosselut, A., Clark, R., Cohan, A., Foerster, J. N., Gal, Y., Hale, S. A., Raji, I. D., Summerfield, C., Torr, P., Ududec, C., Rocher, L., Mahdi, A. Measuring what Matters: Construct Validity in Large Language Model Benchmarks. NeurIPS 2025.

<https://arxiv.org/abs/2511.04703>

Liu, G., Shi, B., Caciularu, A., Szpektor, I., Cohan, A. MDCure: A Scalable Pipeline for Multi-Document Instruction-Following. ACL Main 2025.

<https://aclanthology.org/2025.acl-long.1418>

Lee, H. H., **Liu, G.**, Chen, Y. C., Yeh, S. L. Exploring Quantitative Measures in Metacognition of Emotion. Scientific Reports 2024.

<https://rdcu.be/dwIct>

Liu, G. Perspectives on the Social Impacts of Reinforcement Learning with Human Feedback. 2023.

<https://arxiv.org/abs/2303.02891>

Liu, G. A Robot Rights Curriculum Informed by Western and Eastern Principles. MIT Social and Ethical Responsibilities of Computing Symposium 2023.

<https://robotrights.webflow.io/>

Schaeffer, R., **Liu, G.**, Du, Y., Linderman, S., & Fiete, I. R. Streaming Inference for Infinite Non-Stationary Clustering. ICLR Workshop on Agent Learning in Open-Endedness 2022 & Conference on Lifelong Learning Agents 2022.

<https://arxiv.org/abs/2205.01212>

Schaeffer, R., Du, Y., **Liu, G.**, & Fiete, I. Streaming Inference for Infinite Feature Models. ICML 2022.

<https://proceedings.mlr.press/v162/schaeffer22a.html>

Lee, H. H., **Liu, G.**, Yeh, S. L. I know I'm happy, and I'm right: Metacognition of emotion. European Conference on Visual Perception 2021.

https://journals.sagepub.com/toc/peca/50/1_suppl

Liu, G. Weight Friction: A Simple Method to Overcome Catastrophic Forgetting and Enable Continual Learning. 2019.

<https://arxiv.org/abs/1908.01052>

Liu, G. Evaluating Gammatone Frequency Cepstral Coefficients with Neural Networks for Emotion Recognition from Speech. 2018.

<https://arxiv.org/abs/1806.09010>

Academic Service

AAAI 2026, AAAI-AIA 2026, NeurIPS 2025, ICML 2025, ICLR 2025, AAAI 2025, LLMAgents @ ICLR 2024, ICML 2024, IJCAI 2024, JASA-EL 2023, NeurIPS 2023, MIT Committee on Curricula 2022-2023, NeurIPS 2021, NeurIPS 2020, ICML 2020

Teaching

Head Teaching Assistant, CPSC 4770/5770 Large Language Models, Spring 2026

Department of Computer Science, Yale University, New Haven, CT

Head Teaching Assistant, CPSC 477/577 Natural Language Processing, Spring 2025

Department of Computer Science, Yale University, New Haven, CT

Instructor, Fundamentals of Group Theory, Spring 2021, Spring 2022, Spring 2023

MIT PRIMES Circle, Department of Mathematics, MIT, Cambridge, MA

Lab Assistant, 6.036 Introduction to Machine Learning, Spring 2021

Department of Electrical Engineering & Computer Science, MIT, Cambridge, MA

Teaching Assistant, Fundamentals of Scientific Writing, Summer 2019

Research Science Institute (RSI), MIT, Cambridge, MA

Grader, 18.02 Multivariable Calculus, Spring 2023
Department of Mathematics, MIT, Cambridge, MA

Honors & Awards	Graduate Research Fellowship, National Science Foundation 2023-2028 Teaching and Learning Award, MIT Department of Mathematics 2023 \$5,000 Prize, Envisioning the Future of Computing, MIT Schwarzman College of Computing 2023 Fung Scholar, Fung Foundation 2022 Eloranta Fellow, MIT 2022 Women and Mathematics Program, Institute for Advanced Study 2022 SERC Scholar, MIT Schwarzman College of Computing 2021 & 2022 Undergraduate Academic Award, MIT Department of Brain and Cognitive Sciences 2022 & 2023 MIT International Science and Technology Initiatives Award 2020-2021, 2022, 2023 First Place, Systems Software, Intel International Science and Engineering Fair (ISEF) 2019 \$25,000 Winner, Top 40 Finalist, Regeneron Science Talent Search (STS) 2019 \$25,000 Winner, Third Place, Siemens Competition 2017 Research Science Institute Scholar (RSI) 2018 Community Innovation Award, Society for Science 2018 Intel Excellence in Computer Science Award 2016-2019
Outreach & Activities	Mentorship Program, Women and Gender Minorities in Science at Yale , 2023-2026 Facilitating the professional success of young women and gender minorities in STEM Editor-in-Chief, MIT Undergraduate Research Journal , 2021 & 2022 MIT's only peer-reviewed scientific journal serving the undergraduate population Mentorship Program Director, Women in EECS , 2021-2022 A community for women in EECS that supports, encourages, and empowers them to succeed Resources Chair, MITxHarvard Women in AI , 2021-2022 A supportive community for cisgender women, transgender women, genderqueer individuals, and non-binary individuals pursuing studies or research in AI and machine learning at MIT and Harvard Representative, Dean's Action Group on Social & Ethical Responsibilities of Computing , 2020 Action group to create pedagogical materials for use across all levels of instruction, in order facilitate the development of responsible "habits of mind and action" for those who create and deploy computing technologies, and the creation of technologies in the public interest Vice President of Operations, Delta Phi Epsilon , 2021-2022 International sorority at MIT whose mission is to equip members with the leadership skills to create positive change in the community EMT, MIT Emergency Medical Service , 2020-2021 Organization that provides exception emergency medical care and education to MIT and the surrounding community
Internships & Programs	Summer Camp for Applied Language Exploration (SCALE), HLTCOE, JHU, Summer 2025 MISTI-Taiwan, MIT, Summer 2023 Women and Mathematics Program, Institute for Advanced Study (IAS), Summer 2022 MISTI-Singapore, MIT, Summer 2022 MISTI-Taiwan, MIT, Summer 2020-Summer 2021 Canada/USA Mathcamp, Summer 2019 Research Science Institute, CEE, Summer 2018 Canada/USA Mathcamp, Summer 2017 TN Governor's School for Computational Physics, Summer 2017 MathILy, Summer 2016 MathPath, Summer 2015
Pre-Graduate Projects	Trustworthy Collaborative AI , Prof. Harold Soh, CLeAR Lab, Summer 2022 <i>Department of Computer Science, National University of Singapore, Queenstown, Singapore</i>

Computation and Learning with Brain Assemblies, Prof. Tomaso Poggio, Projects in the Science of Intelligence, Spring 2022

Department of Brain and Cognitive Sciences, MIT, Cambridge, MA

Efficient Streaming Inference for Infinite Nonparametric Models, Prof. Ila Fiete, Fiete Lab, Fall 2021-Spring 2022

Department of Brain and Cognitive Sciences, MIT, Cambridge, MA

Metacognition of Emotion, Prof. Su-Ling Yeh, EPA Lab, Summer 2020-Summer 2021

Department of Psychology, National Taiwan University, Taipei, Taiwan

Lifelong and Meta-Reinforcement Learning for Structured Action Spaces, Prof. Josh Tenenbaum, CoCoSci Lab, Fall 2019-Spring 2020

Department of Brain and Cognitive Sciences, MIT, Cambridge, MA

A Mathematical Framework for Learning Shared Representations for Transfer Learning,

Prof. Lizhong Zheng, Research Science Institute, Summer 2018

Department of Electrical Engineering and Computer Science, MIT, Cambridge, MA

Preventing Domestic Violence Using Emotion Recognition in Speech, 2018

Neural Networks without Multiplications, 2017

Anthropomorphic Facial Emotion Recognition and Generation Objective through Machine Learning, 2017

Recognizing Emotions Using a Physiologically Based Facial Landmark Detection Model and Machine Learning, 2016

Technical Skills

LLM Skills

Designing & analyzing LLM benchmarks, Synthetic data generation & filtering for SFT/RL, End-to-end post-training pipeline development (SFT, DPO/PPO/variants, RM), Designing & training reward models, Rubric-based reward modeling, Large-scale LLM inference, Data efficient training, Training data selection, Distributed LLM inference & training, Training hyperparameter optimization, Experiment tracking, Model versioning and artifact management, Checkpoint selection, LLM meta-evaluation, LLM uncertainty quantification, Designing human data annotation, LLM confidence calibration, LLM faithfulness, Analysis of LLM abstention and hallucination, LLM API integration, Retrieval Augmented Generation (RAG), RAG evaluation, LLM-as-a-Judge prompt & rubric design, Multi-document processing, Multi-document QA & summarization, Multi-hop reasoning, Multi-hop QA, Unanswerable QA evaluation, Scientific QA

Languages

Python, Julia, SQL, Linux, Java, MATLAB, R, Fortran, Rust, LaTeX

Platforms

Google Cloud Platform, AWS, HuggingFace, GitHub, GitLab, Google Colab, JupyterLab, slurm, docker, Jira, Confluence, OpenAI API, Together API, Cohere API, Google AI Studio, Gemini API, HuggingFace

Libraries

PyTorch, trainer, transformers, trl, accelerate, deepspeed, unsloth, bitsandbytes, peft, wandb, TensorFlow, TensorBoard, Keras, OpenCV, dlib, scikit-learn, scipy, numpy, matplotlib, pandas, seaborn, Pyro, multiprocessing, LlamaIndex, ragas, LiteLLM, LangChain, vLLM